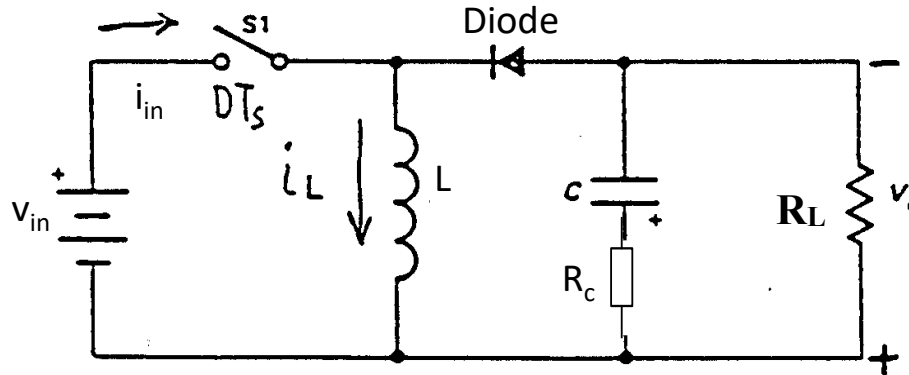


ECE 5254 - Homework #1

Average Model for Buck-Boost Converter

Buck-Boost converter to be analyzed (Switch (S_1) & Diode are IDEAL):



- 1) (40%) Derive the general expressions of (20%) $G_{vd}(s) = v_o(s)/d(s)$, (10%) audio susceptibility $G_{vin2vo}(s) = v_o(s)/v_{in}(s)$, and (10%) Output Impedance $Z_o(s) = v_o(s)/i_o(s)$ for the Buck-boost converter running at continuous current mode. Use the average model concept.

[Notes: (1) For each transfer function, get the small signal equivalent circuit first, then derive the equations from there (show all necessary steps) (2) All transfer functions should be written as the following standard form (see the example for G_{vd} given below) to get full points (3) Include the ESR of the capacitor, R_c]

$$G_{vd}(s) = G_{d0} \frac{[1 + s / \omega_{esr}] \cdot [1 - s / \omega_{rhp}]}{1 + s / (\omega_0 Q) + (s / \omega_0)^2}$$

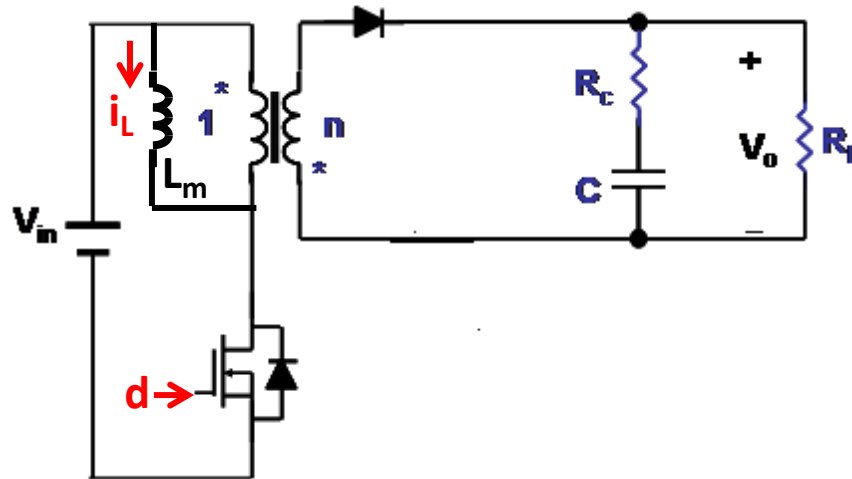
- 2) (20%) For the given circuit parameters: $V_{in} = 12\text{ V}$, $V_o = 4\text{ V}$, $L = 3\text{ uH}$, $C = 300\text{ uF}$, $R_c = 3\text{ m}\Omega$, $F_{sw} = 500\text{ kHz}$, $R_L = 200\text{ m}\Omega$. Plot the derived transfer functions of $G_{vd}(s)$, $G_{vin2vo}(s)$, and $Z_o(s)$ from question 1 (10%) and verify $G_{vd}(s)$, $G_{vin2vo}(s)$, and $Z_o(s)$ with SIMPLIS simulation (10%).

[Note-1: You should plot the bode plot in the frequency range from 100Hz to 500kHz. You must plot the model result and SIMPLIS simulation result in the same bode plot (on top of each other) to see whether the two agree with each other by using MathCAD or MATLAB. Please do not compare them side by side. An example MATLAB file for buck converter is uploaded with this assignment to help you with the plotting. You can modify this file to generate the comparison plots for the buck-boost converter and required setup in question 2.]

[Note-2: Two SIMPLIS simulation files (Buck_Vc2Vo and Buck_Zo) are uploaded with this assignment on canvas. You can modify the schematic and circuit parameters for buck-boost converter and required setup in question 2.]

3) (10%) Derive the general expression of $G_{vd}(s) = v_o(s)/d(s)$ of the Flyback converter operating at continuous current mode. You can transfer the original flyback circuit into equivalent buck-boost circuit first. Then you can derive the transfer function from the equivalent buck-boost circuit. The Flyback converter to be analyzed is given below:

[Hint: You do not need to derive the transfer functions again. You just need to find an equivalence between flyback and buck-boost converters and extend results from question-1]



4) (30%) Discussion.

You must discuss the following (**discuss in detail**):

- (1) Qualitatively explain the conditions under which the expression $G_{vd}(s)$ derived in Question 1 is valid? For the inductance, input voltage, output voltage and switching frequency given in Question 2, find the load resistance range over which expression for $G_{vd}(s)$ in question 1 is valid.
- (2) Does the right half plane zero exist in all transfer functions that you have derived in Question 1 for Buck-boost converter? Explain the physical origins of right-half plane zero in boost and buck-boost converters? Explain the gain and phase characteristic of right half plane zero?

- (3) Why the double pole location is a function of duty cycle in the buck-boost converter?
- (4) What is the effect of circuit parameters V_{in} , R_L , R_c and F_{sw} on transfer function $G_{vd}(s)$ for buck-boost in question 1?
- (5) Explain how the $G_{vd}(s)=v_o(s)/d(s)$ of buck-boost and fly back converters differ? Does the turn ratio have any impact on right-half plane zero (RHPZ) of flyback converter, if so, explain how it impacts RHPZ?
- (6) Explain the average modeling concept and its assumptions introduced in Chapter 1? Why does the small-signal model obtained using the average concept not agree with the simulation around switching frequency?

Due time: Sept. 26, 2024 11:30pm. All the homework should be submitted into canvas before the due time. Either .doc file or .pdf file is accepted. Please change your file name to HW#_First name _Last name for easy recognition. (For example, HW1_Sundaramoorthy_Sridhar.doc or HW1_Sundaramoorthy_Sridhar.pdf)

Late homework should be sent into TA's email sundaramoorthy@vt.edu (10% penalty per day after due time will be strictly executed unless prior arrangements have been made with the instructor).

All the work MUST be your own; the VT honor code will be strictly enforced.